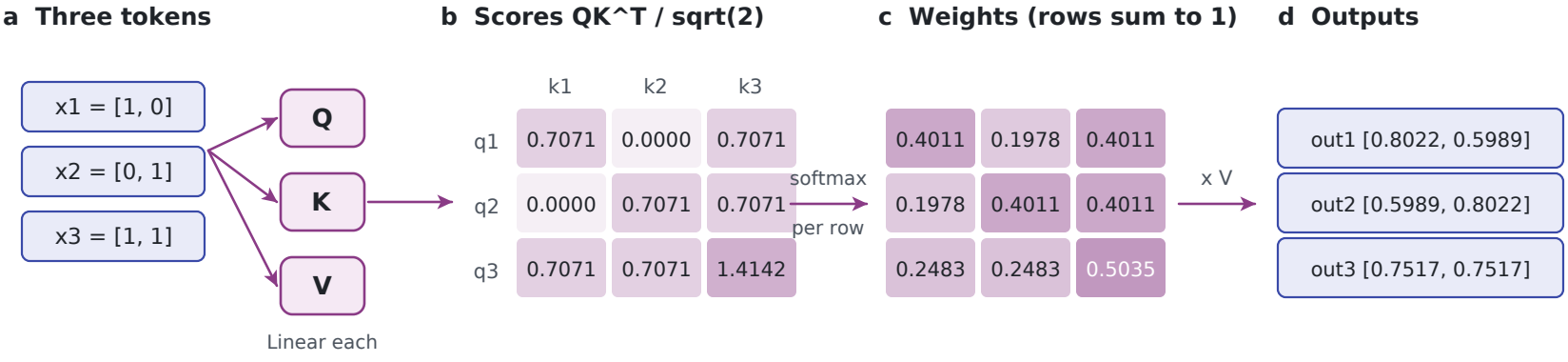

identity projections, so Q = K = V = x (part 12)

out = softmax(Q K^T / sqrt(d)) V

Each output is a blend of all values, weighted by how well its query matches each key.

The score matrix is N x N: at stage 1, N = 1,024 tokens gives 1,048,576 scores per head - the cost part 13 removes.